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A Semi-Supervised Semantic Segmentation Method for Remote Sensing Images via Adaptive Thresholding and Feature Fusion

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***Abstract*—With the rapid proliferation of remote sensing imagery, semantic segmentation has become a pivotal technique for extracting geospatial information. However, the high cost of annotating high-resolution remote sensing images severely limits the practicality of supervised learning approaches. To reduce reliance on labeled data, this paper proposes two key enhancements to the UniMatch semi-supervised segmentation framework. First, a Self-Adaptive Thresholding (SAT) mechanism, inspired by FreeMatch, is integrated to dynamically adjust the criteria for pseudo-label selection, thereby improving both pseudo-supervision quality and training efficiency. Second, the decoder architecture is upgraded by embedding an Adaptive Spatial Feature Fusion (ASFF) module into DeepLabV3+, which enhances multi-scale feature representation and boundary awareness. Experiments on the GID-15 dataset demonstrate that the proposed method achieves a mean Intersection over Union (mIoU) of 79.51% using only one-quarter of the labeled data, representing a 2.81 percentage point improvement over the original UniMatch and significantly outperforming other mainstream approaches.**

***Keywords*—Remote sensing images, Semantic segmentation, Semi-supervised learning, Pseudo-labeling**

I. INTRODUCTION

In the era of big data, remote sensing imagery has become an essential means of acquiring surface-level information and is widely applied in fields such as land use monitoring [1], environmental change detection [2], and disaster assessment [3]. The continuous accumulation of high-resolution remote sensing images has spurred rapid advances in image processing and analysis technologies. Among these, image classification serves as a fundamental task for extracting meaningful information from remote sensing data.

Despite the increasing availability of remote sensing imagery, traditional image classification methods still heavily rely on large quantities of manually annotated samples, posing significant challenges for real-world applications. On the one hand, remote sensing images often exhibit complex spatial structures and diverse spectral characteristics, requiring expert knowledge for accurate annotation. On the other hand, the emergence of large-scale datasets exacerbates the contradiction between annotation cost and the limitations of human labor, making the improvement of classification performance under limited labeled data a pressing issue.

In this context, semi-supervised learning (SSL) has demonstrated considerable potential and practical value. By utilizing a small amount of labeled data alongside a large volume of unlabeled data, SSL methods seek to uncover the underlying structure of the data to enhance model generalization. In recent years, advances in deep learning have further propelled the development of SSL, making it an increasingly prominent strategy for addressing annotation scarcity in remote sensing image classification. SSL has emerged as an effective approach for reducing dependency on labeled data while maintaining high classification accuracy.

II. RELATIVE WORK

A. Remote Sensing Image Classification

Remote sensing image classification is a core task in Earth observation and geospatial information analysis. Early classification methods primarily relied on traditional machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests [4]. These approaches depend heavily on handcrafted features and can perform reasonably well in specific scenarios. However, their generalization capabilities are limited, making them less effective in dealing with complex scenes and high-dimensional data.

With the emergence of deep learning, Convolutional Neural Networks (CNNs) have become the dominant paradigm in remote sensing image classification due to their strong feature representation capabilities. Architectures such as SegNet [5], U-Net [6], and DeepLab [7] have significantly improved pixel-level classification accuracy. More recently, Transformer-based models have been introduced to capture long-range dependencies, further enhancing classification performance.

Nevertheless, the high spectral dimensionality, large image sizes, complex backgrounds, and class imbalance inherent in remote sensing images impose greater demands on model learning. These factors also lead to a strong reliance on large volumes of high-quality annotated data. In semantic segmentation tasks, which require pixel-level annotations, the labeling process becomes especially labor-intensive and often requires domain-specific expertise. The high cost of annotation remains a major bottleneck hindering the widespread adoption

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of supervised learning methods. Consequently, improving model performance under limited annotation has become a key research focus, paving the way for the application of semi-supervised learning in remote sensing image classification.

B. Semi-Supervised Learning Methods

Semi-supervised learning (SSL) is a learning paradigm that effectively addresses the issue of limited labeled data. Its core principle is to leverage a small set of labeled samples in conjunction with a large amount of unlabeled data to improve the generalization performance of models. The most prevalent SSL approaches include pseudo-labeling [8], self-supervised learning [9], consistency regularization, and contrastive learning.

Among these, consistency regularization methods have garnered significant attention for their strong performance in image classification tasks. A representative approach, FixMatch [10], applies both "weak" and "strong" augmentations to the same unlabeled sample, using the high-confidence prediction from the weakly augmented version as a pseudo-label to supervise the strongly augmented counterpart. This technique yields robust and stable pseudo-supervision. Additionally, methods such as Mean Teacher [11], Virtual Adversarial Training (VAT) [12], and ReMixMatch [13] have further improved pseudo-label quality and robustness from various perspectives.

In the context of semantic segmentation, the output is a structured, pixel-level prediction, which introduces additional challenges compared to image classification. Pseudo-labels are particularly prone to errors around object boundaries, and strong augmentations can lead to inconsistencies that impede training. As a result, recent research has sought to extend consistency regularization from the image level to pixel- or feature-level frameworks, aiming to enhance pseudo-supervision accuracy and model stability in segmentation tasks.

III. DATA AND METHODS

A. DATASET

This study utilizes the Gaofen Image Dataset – 15-class fine-grained subset (GID-15), collected by the Chinese Gaofen-2 remote sensing satellite. It is one of the most representative large-scale, high-resolution datasets for land-cover classification in remote sensing.

The GID dataset, developed and annotated by Tong[14], consists of 150 high-quality remote sensing images with a spatial resolution of 4 meters, covering over 50,000 square kilometers across China. Widely used for land-cover classification tasks, the dataset includes imagery from 10 cities and is annotated with 15 semantic classes: industrial land, urban residential, rural residential, traffic land, paddy field, irrigated land, dry cropland, garden plot, arbor woodland, shrub land, natural grassland, artificial grassland, river, lake, and pond.

Following the preprocessing method introduced by Lu [15], the large-scale images were cropped into smaller patches using a sliding window of 512×512 pixels with a stride of 384 pixels. The resulting patches were randomly split into training and test sets in a 4:1 ratio.

B. UniMatchs And Self-Adaptive Thresholding(SAT)

UniMatch [16] is a recently proposed semi-supervised semantic segmentation framework that extends the concept of perturbation-based learning as introduced in FixMatch. It comprises two core modules: UniPerb, which applies learnable feature-space perturbations (e.g., channel-wise dropout) in combination with traditional image-space augmentations (e.g., CutMix, color jittering), and DusPerb, which introduces dual strong perturbations of the same unlabeled image for dual-branch consistency learning, thereby improving the model's robustness and stability.

Although UniMatch has demonstrated strong generalization across benchmark datasets (Pascal VOC, Cityscapes, COCO), and tasks like change detection and medical image segmentation, it still has limitations. A notable drawback lies in its reliance on a fixed high-confidence threshold (typically 0.95) to select pseudo-labels. During early training stages, model predictions tend to be less confident, resulting in many valuable unlabeled samples being discarded. This reduces the utilization of available data and slows down training convergence. Furthermore, the inherent characteristics of remote sensing imagery—namely large variations in object scale, complex textures, and ambiguous class boundaries—impose higher demands on the quality of pseudo-labels. Under such conditions, the use of fixed, high-confidence thresholds often fails to generate accurate pseudo-labels for unlabeled data, especially for fine-grained categories. This limitation hinders the model's ability to effectively classify subtle land-cover types in remote sensing scenes.

To address this, we incorporate the Self-Adaptive Thresholding (SAT) mechanism proposed in FreeMatch [17] into the UniMatch framework. SAT allows the model to dynamically adjust the pseudo-label selection threshold based on the learning status, thereby enhancing both pseudo-supervision quality and training efficiency.

SAT introduces two complementary components:

1) The global threshold eflects the model's overall prediction confidence on unlabeled data. It is updated using an Exponential Moving Average (EMA) of the maximum confidence values:

$$\tau_t = \lambda\tau_{t-1} + (1-\lambda)\frac{1}{\mu B}\sum_{b=1}^{\mu B}\max(q_b) \tag{1}$$

Where τ_t is the current global threshold, q_b is the softmax output for the b-th unlabeled sample, μB is the number of unlabeled samples in the batch, $\lambda \in [0,1)$ is the EMA decay factor, typically set to 0.999.

2) The class-specific thresholds are designed to reflect the model's prediction status for each class. For every class c, the average confidence across the batch is also updated via EMA:

$$\tilde{p}_t(c) = \lambda\tilde{p}_{t-1}(c) + (1-\lambda)\frac{1}{\mu B}\sum_{b=1}^{\mu B} q_b(c) \tag{2}$$

These class-wise thresholds are then normalized and combined with the global threshold to generate the final class-adaptive threshold used for pseudo-label filtering:

$$\tau_t(c) = \frac{\tilde{p}_t(c)}{\max_{c'}\tilde{p}_t(c')}\tau_t \tag{3}$$

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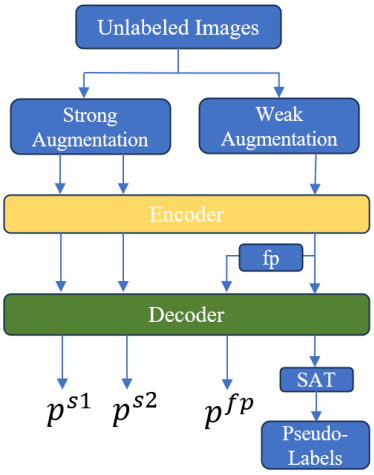


Figure. 1. UniMatch+SAT Architecture

As shown in Figure 1, each unlabeled image is subjected to both weak and strong perturbations. Two independent strong perturbations are applied and fed into the network, resulting in two distinct predictions, p^{s1} and p^{s2}, which differ due to the randomness and independence of the perturbations. Meanwhile, another branch of the same image is weakly augmented and processed by the encoder. The resulting feature is then split into two paths: the first applies a strong feature-level perturbation using Dropout, followed by the decoder to generate the prediction p^{fp} ;the second path bypasses additional perturbation and is passed through the network to obtain a pseudo-label via the SAT module. This pseudo-label is then used as the training target for supervising p^{s1}, p^{s2} and p^{fp} ,thereby enhancing the model's consistency and robustness under different perturbations.

C. Adaptive Spatial Feature Fusion(ASFF)

Although SAT significantly improves pseudo-label utilization, semantic segmentation of remote sensing imagery still faces a critical challenge: ambiguous object boundaries and complex multi-scale target overlaps. Land-cover categories (e.g., water bodies, roads, buildings, and farmlands) in such images often exhibit substantial scale variations, diverse textures, and blurred edges. These characteristics degrade the performance of conventional decoders, especially in accurately delineating object boundaries and segmenting small-scale targets, due to insufficient feature fusion.

To address this limitation, we enhance the decoder of UniMatch by integrating an Adaptive Spatial Feature Fusion (ASFF) module[18], which facilitates more effective fusion between low-level spatial details and high-level semantic representations. The ASFF module employs a learnable weighting mechanism to adaptively adjust the contribution of shallow and deep features at each spatial location. This allows the model to selectively emphasize boundary information or semantic context depending on the scene, thereby improving the segmentation quality, particularly in complex and multi-scale remote sensing scenarios.

As shown in Figure 2. The ASFF module is composed of the following key components. First, the shallow feature map c1 and the deep feature map c4 are individually compressed using 1×1 convolutions to reduce computational complexity. Next, separate weight vectors are predicted for the two compressed features, which are then concatenated and passed through a Softmax function to generate spatially adaptive fusion ratios for each location. These weights are then used to perform a weighted fusion of c1 and c4, and the fused features are subsequently processed by a 3×3 convolution for spatial reconstruction. Finally, the output is fed into the classification head for semantic segmentation prediction.

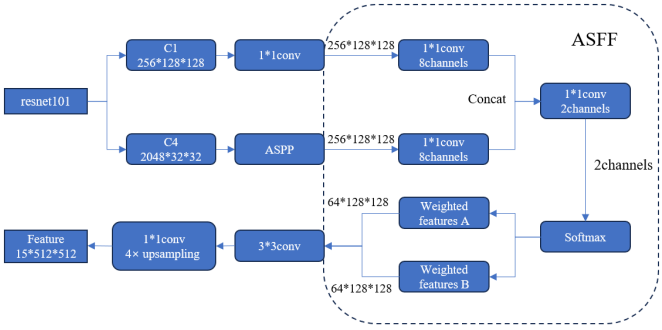


Figure. 2. ASFF Architecture

IV. EXPERIMENTS AND RESULTS

A. Experimental Settings

For semi-supervised learning, we adopt the commonly used DeepLabV3+ architecture with ResNet-101 as the backbone, following standard practice in most SSL-based segmentation studies. In our semi-supervised experimental setup, we designate 1/4 and 1/8 of the dataset as labeled data, respectively. Each mini-batch consists of 4 labeled and 4 unlabeled samples.

All training processes are conducted on an NVIDIA A100 GPU server. Training is performed using the Stochastic Gradient Descent optimizer with a momentum of 0.9 and a weight decay of 1e-4. The initial learning rate is set to 0.01 and is dynamically adjusted using a polynomial decay strategy. The model is trained for a total of 80 epochs, and the cross-entropy loss is used as the loss function.

B. Evaluation Metrics

To comprehensively evaluate the performance of the semi-supervised segmentation models, we adopt Mean Intersection over Union (mIoU) as the evaluation metric. For a given class c out of a total of N classes, the mIoU is defined as follows:

$$IoU_c = \frac{TP_C}{TP_C+FP_C+FN_C} \tag{4}$$

$$mIoU = \frac{1}{N}\sum_{c=1}^{N} IoU_c \tag{5}$$

C. Results And Visualization

We compare our improved method with several baseline and state-of-the-art approaches on the GID-15 dataset under identical hyperparameter settings. The evaluated methods include the baseline models LSST and AP-Semi, as well as classic semi-supervised approaches such as FixMatch, UniMatch, and UniMatch enhanced with either the ASFF or SAT module. Table 1 reports the segmentation accuracy in terms of mIoU, achieved by each method using 1/4 and 1/8 fractions of labeled data, respectively.

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TABLE I. DIFFERENT METHODS OF MIOU

Model	1/4	1/8
LSST[15]	71.28	66.38
AP-Semi[19]	73.59	/
FixMatch	74.56	68.25
UniMatch	76.70	71.05
UniMatch+ASFF	76.10	70.55
UniMatch+SAT	78.36	71.57
UniMatch+SAT+ASFF	**79.51**	**72.16**

As shown in Table I, under the 1/4 labeled data setting, baseline methods LSST and AP-Semi achieved mIoU scores of 71.28% and 73.59%, respectively. FixMatch demonstrated superior performance with 74.56%, while UniMatch attained 76.70%, showcasing its effectiveness in leveraging dual-stream perturbations and consistent pseudo-labeling. The integration of the SAT mechanism yielded a significant improvement: UniMatch+SAT achieved 78.36% mIoU, representing a 1.66 percentage-point gain over vanilla UniMatch. Although UniMatch+ASFF produced a marginally lower result 76.10%, our combined approach UniMatch+SAT+ASFF attained the highest mIoU of 79.51%, demonstrating the synergistic effects of adaptive thresholding and multi-scale feature fusion.

Under the more challenging 1/8 labeled data condition, all methods exhibited expected performance degradation. LSST declined to 66.38%, while FixMatch dropped to 68.25%. UniMatch maintained robust performance at 71.05%, confirming its resilience under label scarcity. The SAT module again proved beneficial, with UniMatch+SAT reaching 71.57%. Further enhancement was achieved through ASFF integration, establishing a new state-of-the-art mIoU of 72.16%. These results substantiate the efficacy of our proposed enhancements in extreme low-annotation scenarios, particularly highlighting the complementary contributions of SAT and ASFF.

As shown in Table II, which presents the per-class IoU results under the 1/4 labeled data setting for FixMatch, UniMatch, and UniMatch+SAT+ASFF across all fifteen categories, UniMatch consistently outperforms FixMatch in most classes, confirming the effectiveness of its extended perturbation strategy and pseudo-label design. Although UniMatch and UniMatch+SAT+ASFF exhibit similar performance across most categories, the UniMatch+SAT+ASFF method achieves significant improvements in small-scale categories such as Shrub land and Artificial grassland, indicating its enhanced capability in recognizing fine-grained, small-scale objects. It delivers more balanced and robust performance across various land-cover types, demonstrating strong effectiveness in multi-scale, edge-sensitive, and class-imbalanced scenarios.

TABLE II. VARIOUS CLASSES OF IOU UNDER 1/4 LABELED DATA

Label	FixMatch	Unimatch	UniMatch+ SAT+ASFF
Industrial land	72.52	75.46	**76.27**
Urban residential	79.57	**81.91**	80.93
Rural residential	72.10	**74.18**	74.00
Traffic land	73.06	77.82	**78.33**
Paddy field	80.92	84.49	**85.14**
Irrigated land	90.98	92.33	**93.02**
Dry cropland	79.39	**80.48**	80.11
Garden plot	35.89	**44.48**	40.39
Arbor woodland	98.29	**98.97**	98.64
Shrub land	29.15	31.59	**68.84**
Natural grassland	90.35	**94.96**	92.97
Artificial grassland	64.97	61.12	**67.70**
River	90.79	93.41	**94.03**
Lake	85.22	83.95	**85.49**
Pond	75.25	75.42	**76.82**

Notably, under the 1/4 labeled data configuration, significant variations in training efficiency emerge among the compared methods. When employing the original DeepLabV3+ as the decoder architecture, FixMatch demonstrates a processing time of approximately 317 seconds per training epoch, while UniMatch requires about 304 seconds per epoch. Our proposed method achieves enhanced computational efficiency, completing each epoch in just 267 seconds.

Furthermore, our architectural modifications yield additional benefits in model compactness. Although the newly introduced ASFF module adopts a more structurally sophisticated design, it replaces the original fusion module's two large-channel 3×3 convolutions with several lightweight 1×1 convolutions of reduced channel size. As a result, the overall parameter count is lower. This structural refinement reduces the model size from UniMatch's 454 MB to 445 MB in our implementation. These improvements collectively demonstrate the effectiveness of our approach in optimizing both computational efficiency and model storage requirements.

To provide a more intuitive comparison of the segmentation performance across different methods and scenes, Figure 3 presents visualized prediction results for several representative test images.

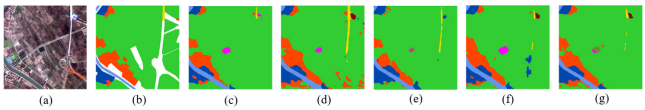


Figure. 3. Visualization of Prediction Results(a)input image. (b)label. (c)FixMatch.(d)UniMatch. (e)UniMatch+ASFF. (f) UniMatch+SAT. (g) UniMatch+ASFF+SAT.

V. CONCLUSION

In this work, we propose a refined semi-supervised semantic segmentation framework tailored for high-resolution remote sensing imagery, building upon and extending the UniMatch architecture. To address limitations of static thresholding in pseudo-label generation, we integrate the SAT mechanism, which dynamically adjusts confidence thresholds based on model training status. This adaptation significantly improves the quality of pseudo-labels and training efficiency on remote sensing images.

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Furthermore, to better handle the multi-scale nature and boundary complexity of land-cover classes, we introduce an ASFF module into the decoder architecture. This module enables spatially adaptive feature integration between low- and high-level features, leading to improved segmentation of small or edge-sensitive objects.

Extensive experiments on the GID-15 dataset demonstrate the superiority of our proposed approach. The combined UniMatch+SAT+ASFF model achieves a state-of-the-art mIoU of 79.51% using only 1/4 of the labeled data, outperforming baseline methods and demonstrating enhanced robustness in both overall performance and class-wise segmentation. Even under the more challenging 1/8 label setting, our method maintains strong generalization.

Although our method demonstrates overall improvements in mIoU, there is a notable performance degradation in the Garden plot category, likely due to the sparsity of this class in the dataset. Furthermore, we have not explicitly evaluated the method under extreme class imbalance or image degradation scenarios, such as cloud occlusion or resolution loss. Addressing these issues in future work will further strengthen the robustness of our framework.

Overall, this study demonstrates the promise of combining adaptive pseudo-label selection and spatial feature fusion for semi-supervised segmentation of remote sensing imagery, providing a scalable and efficient solution in label-scarce environments.

REFERENCES

[1] Ahmadlou M, Karimi M, Sammen S S, Alsafadi K. Three novel cost-sensitive machine learning models for urban growth modelling[J]. Geocarto International, 2024, 39(1): 2353252.

[2] Hemati M A, Hasanlou M, Mahdianpari M, Mohammadimanesh F. A systematic review of landsat data for change detection applications: 50 years of monitoring the earth[J]. Remote sensing, 2021, 13(15): 2869.

[3] Zheng Z, Zhong Y, Wang J, Ma A, Zhang L. Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters[J]. Remote Sensing of Environment, 2021, 265: 112636.

[4] Khan A, Sudheer M. Machine learning-based monitoring and modeling for spatio-temporal urban growth of Islamabad[J]. The Egyptian Journal of Remote Sensing and Space Science, 2022, 25(2): 541-550.

[5] Jonnalagadda A V, Hashim H A. SegNet: A segmented deep learning based Convolutional Neural Network approach for drones wildfire detection[J]. Remote Sensing Applications: Society and Environment, 2024, 34: 101181.

[6] Azad R, Aghdam E K, Rauland A, et al. Medical image segmentation review: The success of u-net[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2024.

[7] Wang J, Liu X. Medical image recognition and segmentation of pathological slices of gastric cancer based on Deeplab v3+ neural network[J]. Computer methods and programs in biomedicine, 2021, 207: 106210.

[8] Arazo E, Ortego D, Albert P, et al. Pseudo-labeling and confirmation bias in deep semi-supervised learning[C]//2020 International joint conference on neural networks (IJCNN). IEEE, 2020: 1-8.

[9] Chen T, Kornblith S, Norouzi M, Hinton G. A simple framework for contrastive learning of visual representations[C]//International conference on machine learning. PmLR, 2020: 1597-1607.

[10] Sohn K, Berthelot D, Carlini N, et al. Fixmatch: Simplifying semi-supervised learning with consistency and confidence[J]. Advances in neural information processing systems, 2020, 33: 596-608.

[11] Döbler M, Marsden R A, Yang B. Robust mean teacher for continual and gradual test-time adaptation[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023: 7704-7714.

[12] Li Y, Liu J, Jiang Y, Liu Y, Lei B. Virtual adversarial training-based deep feature aggregation network from dynamic effective connectivity for MCI identification[J]. IEEE transactions on medical imaging, 2021, 41(1): 237-251.

[13] Berthelot D, Carlini N, Cubuk E D, Kurakin A. Remixmatch: Semi-supervised learning with distribution alignment and augmentation anchoring[J]. arXiv preprint arXiv:1911.09785, 2019.

[14] Tong X Y, Xia G S, Lu Q, et al. Land-cover classification with high-resolution remote sensing images using transferable deep models[J]. Remote Sensing of Environment, 2020, 237: 111322.

[15] Lu X, Jiao L, Liu F, et al. Simple and efficient: A semisupervised learning framework for remote sensing image semantic segmentation[J]. IEEE Transactions on Geoscience and Remote Sensing, 2022, 60: 1-16.

[16] Yang L, Qi L, Feng L, et al. Revisiting weak-to-strong consistency in semi-supervised semantic segmentation[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2023: 7236-7246.

[17] Wang Y, Chen H, Heng Q, et al. Freematch: Self-adaptive thresholding for semi-supervised learning[J]. arXiv preprint arXiv:2205.07246, 2022.

[18] Liu S, Huang D, Wang Y. Learning spatial fusion for single-shot object detection[J]. arXiv preprint arXiv:1911.09516, 2019.

[19] Bai L, Wang H, Zhang X, Qin W, Liu B, Du S. AP-semi: Improving the semi-supervised semantic segmentation for VHR images through adaptive data augmentation and prototypical sample guidance[J]. IEEE Transactions on Geoscience and Remote Sensing, 2024, 62: 1

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